

$$F(s) = \frac{1.5}{s s^2 + s s + 1} \Rightarrow 2. \text{ řád}$$

$$F(z) = \frac{Y(z)}{U(z)} = \frac{b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}} \Leftrightarrow Y(z) \cdot (1 + a_1 z^{-1} + a_2 z^{-2}) = U(z) \cdot (b_1 z^{-1} + b_2 z^{-2}) \Leftrightarrow$$

$$\Leftrightarrow Y(z) + a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) = b_1 z^{-1} U(z) + b_2 z^{-2} U(z) \Leftrightarrow$$

$$\Leftrightarrow y(k) + a_1 y(k-1) + a_2 y(k-2) = b_1 u(k-1) + b_2 u(k-2) \Leftrightarrow$$

$$\Leftrightarrow y(k) = -a_1 y(k-1) - a_2 y(k-2) + b_1 u(k-1) + b_2 u(k-2)$$